常微分复习

一阶方程

变量分离 $\frac{dy}{dx}=\varphi(x)\phi(y)$ $\begin{cases}\phi(y)\neq0 \Rightarrow \int\frac{dy}{\phi(y)}=\int\varphi(x)dx \\ \phi(y)=0 \Rightarrow \phi(y_0)=0 \quad y=y_0 \text{ 解?}\end{cases}$

变量型齐次 $\frac{dy}{dx}=g(\frac{y}{x})$ 令 $u=\frac{y}{x}$

一阶线性齐次 $\frac{dy}{dx}+P(x)y=0$ $\quad y=ce^{-\int P(x)dx}$

一阶线性非齐次 $\frac{dy}{dx}+P(x)y=f(x)$ $\quad y=e^{-\int P(x)dx}\left[\int f(x)e^{\int P(x)dx}dx+C\right]$

伯努利 $\frac{dy}{dx}+P(x)y=f(x)y^n$ $\begin{cases}y\neq0 \quad \text{令 } z=y^{1-n} \\ y=0\end{cases}$

全微分 $Mdx+Ndy=0$ $\quad \frac{\partial(\mu M)}{\partial y}=\frac{\partial(\mu N)}{\partial x}$ $\quad \frac{dy}{dx}+P(x)y=f(x) \Rightarrow \mu(x)=e^{\int P(x)dx}$

二阶方程

可降解 $y''=f(x)$ 直积2次

$y''=f(x,y')$ 令 $p=y'$ $\quad \frac{dp}{dx}=f(x,p)$

$y''=f(y,y')$ 令 $p=y'$ $\quad p\frac{dp}{dy}=f(y,p)$

二阶常系数线性齐次 $y''+py'+qy=0$ $\quad r^2+pr+q=0$

	通
$r_1\ r_2$	$C_1e^{r_1x}+C_2e^{r_2x}$
r	$C_1e^{rx}+C_2xe^{rx}$
$\alpha\pm i\beta$	$e^{\alpha x}(C_1\cos\beta x+C_2\sin\beta x)$

$(r^2+pr+q)^2=0$

$r_{1,2}\quad \alpha+i\beta$

$r_{3,4}\quad \alpha-i\beta$

$e^{\alpha x}\left[(C_1+C_2x)\cos\beta x+(C_3+C_4x)\sin\beta x\right]$

二阶常系数线性非齐次 $y''+py'+qy=P_m(x)e^{\alpha x}$ $\quad r^2+px+q=0$ $\quad$ 特 $\tilde{y}=Q(x)e^{\alpha x}$ $\quad$ 通 = 齐通 + 特

α为	
非	$Q(x)=R_m(x)$
单	$Q(x)=xR_m(x)$
重	$Q(x)=x^2R_m(x)$

$y''+py'+qy=P_m(y)e^{\alpha x}\cos bx \;/\; P_m(y)e^{\alpha x}\sin bx$ $\quad$ 拆接 → 通 → 特 (求α)

二阶变系数线性 欧拉 $x^ny^{(n)}+a_1x^{n-1}y^{(n-1)}+\dots+a_{n-1}xy'+a_ny=f(x)$ $\quad f(x)$连续

$\begin{cases}\text{换元 } x=e^t \\ \text{算子法 } D=\frac{d}{dt}\end{cases}$

$xy'=Dy$

$x^2y''=D(D-1)y$

$x^3y'''=D(D-1)(D-3)y$

齐次 - Liouville $y''+P(x)y'+Q(x)y=0$ $\quad$ 猜 y_1 $\quad y=y_1\left[C_1\int|y_1|^{-2}e^{-\int P(x)dx}dx+C_2\right]$

非齐次 - 常数变易 $y''+P(x)y'+Q(x)y=f(x)$

已知齐次通 $C_1y_1+C_2y_2$ 求非齐特 $\tilde{y}=u_1y_1+u_2y_2$ $\quad \begin{cases}u_1'y_1+u_2'y_2=0 \\ u_1'y_1'+u_2'y_2'=f(x)\end{cases}$

常系数线性方程组

齐/非齐次 — 消元转二阶方程

一阶齐次 — 特征向量法 $\begin{cases}\frac{dx}{dt}=a_{11}x+a_{12}y \\ \frac{dy}{dt}=a_{21}x+a_{22}y\end{cases}$ $\quad \begin{cases}x=k_1e^{\lambda t} \\ y=k_2e^{\lambda t}\end{cases}$ $\quad \lambda\begin{bmatrix}k_1\\k_2\end{bmatrix}=\underbrace{\begin{bmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{bmatrix}}_{=A}\begin{bmatrix}k_1\\k_2\end{bmatrix}$

A 的特值 $\lambda_1\ \lambda_2$ — 无关特向 $\vec{v_1}\ \vec{v_2}$

特 $\quad \vec{x_1}=v_1e^{\lambda_1 t} \quad \vec{x_2}=v_2e^{\lambda_2 t}$

通 $\quad \vec{x}=C_1x_1+C_2x_2$ 可推广

复 λ 与 v $\quad A$ → 特 → 实虚分离 → 实特 → 通

一阶非齐次 — 常数变异法 $\begin{cases}\frac{dx}{dt}=a_{11}x+a_{12}y+f_1(t) \\ \frac{dy}{dt}=a_{21}x+a_{22}y+f_2(t)\end{cases}$

$\frac{d}{dt}(X\vec{C}(t))=X'\vec{C}(t)+X\vec{C}'(t)=AX\vec{C}(t)+\vec{f}(t)$

$\Rightarrow X\vec{C}'(t)=\vec{f}(t) \quad \Rightarrow \vec{C}'(t)=X^{-1}\vec{f}(t)$

$\frac{d}{dt}\vec{x}=A\vec{x}$ $\quad$ 通：$\vec{x}=C_1\vec{x_1}+C_2\vec{x_2}$

$\frac{d}{dt}\vec{x}=A\vec{x}+\vec{f}(t)$ $\quad$ 特：$\vec{x}^*=C_1(t)\vec{x_1}+C_2(t)\vec{x_2}$

$\frac{d}{dt}\vec{x}=A\vec{x}+\vec{f}(t)$ $\quad$ 通：$\vec{x}=C_1\vec{x_1}+C_2\vec{x_2}+\vec{x}^*$

积分

$\int x^{\alpha} dx = \frac{1}{\alpha+1} x^{\alpha+1} + C$

$\int a^x dx = \frac{a^x}{\ln a} + C$

$\int \tan x \, dx = -\ln|\cos x| + C$

$\int \sec^2 x \, dx = \tan x + C$

$\int \sec x \tan x \, dx = \sec x + C$

$\int \sec x \, dx = \ln|\sec x + \tan x| + C$

$\int \frac{1}{x} dx = \ln|x| + C$

$\int \ln x \, dx = x\ln x - x + C$

$\int \cot x \, dx = \ln|\sin x| + C$

$\int \csc^2 x \, dx = -\cot x + C$

$\int \csc x \cot x \, dx = -\csc x + C$

$\int \csc x \, dx = \ln|\csc x - \cot x| + C$

$dx = \frac{1}{a} d(ax+b)$

$x dx = -\frac{1}{2} d(a^2 - x^2) = \frac{1}{2} d(x^2 - a^2)$

$\frac{1}{\sqrt{x}} dx = 2d\sqrt{x}$

$\frac{1}{x} dx = d\ln|x|$

$e^x dx = de^x$

$\cos x \, dx = d\sin x$

$\sin x \, dx = -d\cos x$

$\frac{1}{\sqrt{1-x^2}} dx = d\arcsin x$

$\frac{1}{1+x^2} dx = d\arctan x$

$\frac{x}{\sqrt{1-x^2}} dx = d\sqrt{1-x^2}$

$\frac{x}{\sqrt{1+x^2}} dx = d\sqrt{1+x^2}$

$\int \sqrt{a^2+x^2} dx = \frac{1}{2}\left(x\sqrt{a^2+x^2} + a^2 \ln\left|x + \sqrt{x^2+a^2}\right|\right) + C$

$\int \sqrt{x^2-a^2} dx = \frac{1}{2}\left(x\sqrt{x^2-a^2} - a^2 \ln\left|x + \sqrt{x^2-a^2}\right|\right) + C$

$\int \sqrt{a^2-x^2} dx = \frac{1}{2}\left(a^2 \arcsin\frac{x}{a} + x\sqrt{a^2-x^2}\right) + C$

$\int \frac{1}{\sqrt{a^2+x^2}} dx = \ln\left|\sqrt{x^2+a^2} - x\right| + C$

$\int \frac{1}{\sqrt{x^2-a^2}} dx = \ln\left|\sqrt{x^2-a^2} - x\right| + C$

$\int \frac{1}{\sqrt{a^2-x^2}} dx = \arcsin ax + C$

$\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \arctan\frac{x}{a} + C$

$\int \frac{1}{x^2-a^2} dx = \frac{1}{2a} \ln\left|\frac{x-a}{x+a}\right| + C$

$\int \frac{1}{a^2+x^2} dx = \frac{1}{2a} \ln\left|\frac{a+x}{a-x}\right| + C$

$\int \frac{1}{(a^2+x^2)^{\frac{3}{2}}} dx = \frac{x}{a^2\sqrt{a^2+x^2}} + C$

$\int \frac{dx}{x(1+x^2)} = \int \frac{dx^2}{2(1+x^2)}$